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Preliminary paleomagnetic data from the Sierra Grande Formation: Tectonic consequences of the first mid–Paleozoic paleopoles from Patagonia

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Abstract—The first two mid–Paleozoic paleomagnetic poles from Patagonia were obtained from a preliminary paleomagnetic study of the Sierra Grande Formation, exposed at 65.4°W, 41.6°S in Argentina. The first pole, SG1, is located at 238.0°E, 3.4°N (N = 14, D1 = 14.3°, D2 = 8.5°) and corresponds to a ferriferous horizon intercalated in the lower member of the formation, which is assigned to the Wenlockian. The second pole, SG2, located at 283.5°E, 42.0°S (N = 10, D1 = 20.4°, D2 = 20.2°), belongs to the upper member of the formation and is assigned to the Late Silurian–Early Devonian. These pole positions are interpreted according to different tectonic hypotheses. The position of SG1 is more easily understood if Patagonia had an allochthonous origin, for it does not agree with the presently most accepted Silurian pole position of Gondwana. Nevertheless, the position of SG2, consistent with the Late Silurian–Early Devonian Gondwana poles, suggests that Patagonia was next to collide with South America in the Early Devonian. On the other hand, the hypothesis of an autochthonous Patagonia is supported by the consistent position of SG2 with respect to the coetaneous poles of Gondwana, while SG1 would indicate that the reversed polarity path of Schmidt and Morris (1977) should be taken as the most probable early-middle Paleozoic apparent polar wander path of Gondwana.

Resumen—Un estudio paleomagnético preliminar de la F. Sierra Grande, aflorante a los 65.4°O, 41.6°S, Argentina, dio como resultado los dos primeros polos paleomagnéticos del Paleozoico Medio de Patagonia. SG1, el polo paleomagnético del horizonte ferrífero Rosales, intercalado en el miembro inferior de la formación y asignado al Wenlockiano, está ubicado en 238.0°E, 3.4°N (N = 14, D1 = 14.3°, D2 = 8.5°), mientras que SG2, correspondiente al miembro superior de la formación, de edad Silúrico tardio–Devónico temprano está situado en 283.5°E, 42.0°S (N = 10, D1 = 20.4°, D2 = 20.2°). Estas posiciones polares han sido interpretadas de acuerdo a dos hipótesis diferentes sobre la evolución tectónica de la Patagonia. La posición de SG1 es facilmente interpretada si se supone un origen alóctono para la Patagonia, ya que no coincide con la posición polar silúrica del Gondwana aparentemente más aceptada actualmente. La posición de SG2, concordante con los polos coetáneos gondwánicos, indica en este caso que para el Devónico temprano la Patagonia se encontraba muy cerca de América del Sur y próxima a su colisión con la misma. Por el contrario, la hipótesis de un origen autóctono para la Patagonia es avalada por la posición de SG2; mientras que SG1 indicaría que la curva de desplazamiento polar aparente, propuesta primeramente por Schmidt y Morris (1977) y basada en adoptar una polaridad opuesta para los polos del Paleozoico Inferior y Medio del Gondwana es la más probable, ya que solo es consistente con la misma.

INTRODUCTION

IN THE FIRST PART of this century, some geologists proposed that the extreme southern part of South America, called "the Patagonian Block," had a geologic and tectonic evolution apart from the South American continent, the Brazilian Craton, during part of the Phanerozoic (Keidel, 1925; Windhausen, 1931). In the past decade, other authors have suggested models for the tectonic evolution of southern South America that include an allochthonous origin for Patagonia. These tectonic models can be divided into two different types. The first proposes a northwestern (present–day coordinates) provenance for Patagonia, with docking mainly by sinistral transcurrence, probably during the late Paleozoic to early Mesozoic (Dalmayrac *et al.*, 1980; Martinez, 1980; Sellés Martinez, 1988). The second suggests a frontal collision of the Patagonian Block with the South American continent (Ramos, 1984; 1986; 1988); this hypothetical collision would have ocurred between the Late Devonian and Late Permian. Nonetheless, many geologists have preferred an "autochthonous hypothesis" — *i.e.*, that Patagonia and the Brazilian Craton have not undergone important relative displacements since at least the mid– or late Paleozoic (Halpern, 1972; Forsythe, 1982; Uliana and Biddle, 1987; among others).

Some paleomagnetic data from late Paleozoic rocks in the Southern Andes outside Patagonia (Vilas and Valencio, 1982) were initially interpreted as possible evidence of a late Paleozoic accretional origin of that region and Patagonia. However, new paleomagnetic data (Vilas and Rapalini, 1989; Bobbio *et al.*, 1990) have ruled out this interpretation. Unexpected warm–climate Permian fossils found in Pata-

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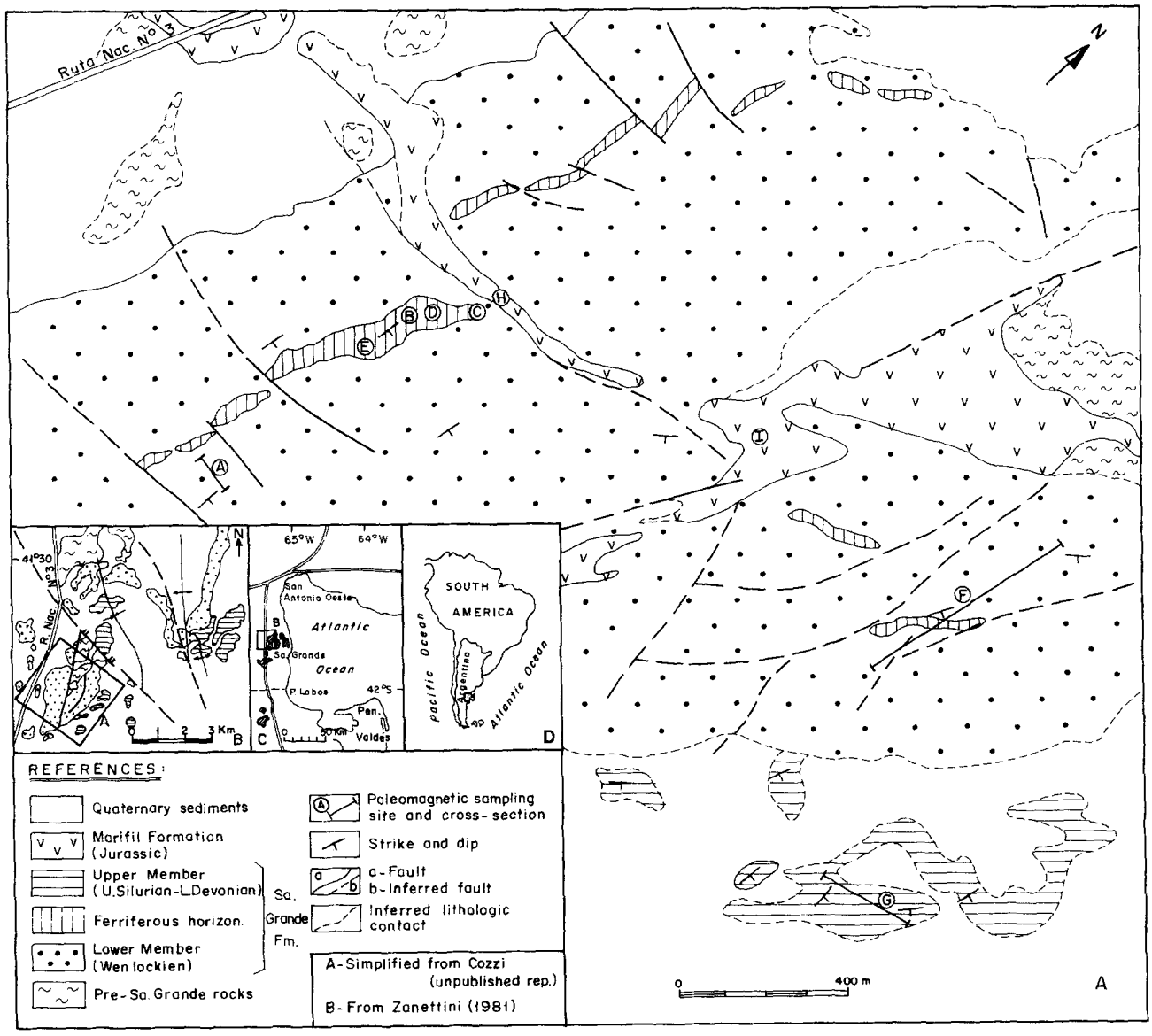


Fig. 1. A, geologic map of the sampling area of the Sierra Grande Formation and the location of the paleomagnetic sampling sites; B, geologic sketch of the Sierra Grande region; C and D, location maps.

gonia (Cúneo and Andreis, 1983) were taken as possible evidence of an allochthonous origin of this continental block, but they can be easily interpreted in terms of climatic warming in all of South America during this period (Sempere, 1989; Rapalini, 1990).

Paleomagnetism has been broadly shown to be a very useful tool in elucidation of the "allochthonous/autochthonous" controversy regarding the origin of terranes, but no paleomagnetic data were available from pre–Late Carboniferous rocks of Patagonia. The acquisition of reliable paleomagnetic poles (PPs) of early and mid–Paleozoic rocks in this continental block is thus very important for better understanding the tectonic evolution of southern South America.

A sequence of Silurian–Devonian marine sedimentary rocks with intercalated ferriferous horizons, the Sierra Grande Formation, is exposed in northeastern Patagonia. A preliminary paleomagnetic sampling of these rocks was carried out several years ago by one of the authors (JFV). The paleomagnetic study was started but, for technical reasons, could not be finished on that occasion. This study was finished recently, with the help of new and repaired instruments, as well as implementation of an integral computing system for paleomagnetism developed by Oviedo (1989). The results of this work, the first mid–Paleozoic paleomagnetic data from Patagonia, are reported here.

GEOLOGY

The Sierra Grande Formation is exposed in northeastern Patagonia near the town of Sierra Grande (41.6°S, 65.4°W; Fig. 1). The formation is a mid–Paleozoic sedimentary sequence that covers the Precambrian–early Paleozoic metamorphic and plutonic rocks constituting the North Patagonian Massif (Caminos and Llambias, 1984). The Punta Sierra granodiorite, which underlies the Sierra Grande Formation, has been K/Ar–dated at 450±20 Ma (Nuñez *et al.*, 1975; Cortés *et al.*, 1984). Because of the economic

importance of ferriferous horizons intercalated in the sedimentary sequence, the stratigraphy of the Sierra Grande Formation has been thoroughly studied by many authors. A complete review of previous work and a description of this formation can be found in Zanettini (1981).

Zanettini (1981) divided the Sierra Grande Formation into two members. The lower, San Carlos Member consists of more than 500 meters of marine quarzites, fine sandstones, siltstones, and shales. A ferriferous horizon, the Rosales horizon, is intercalated at about the middle of the sequence. This horizon, the thickness of which varies between 10 and 30 meters, is composed of oolitic hematite, but magnetite has also been reported (Zanettini, 1981). The upper, Herrada Member, is composed of red to pink and brown coarse, medium, and fine sandstones and some siltstones, and is a minimum of 1000 meters thick. Another ferriferous horizon, the Alfaro horizon, is also intercalated near the bottom of the Herrada Member. This horizon is seldom thicker than 10 meters and is also composed mainly of oolitic hematite.

The age of the Sierra Grande Formation has been determined accurately by studies of its fossils. According to Manceñido and Damborenea (1984), the fossils found in the lower member (Herrada) indicate a Wenlockian age. The age of the upper member (San Carlos) is not as well constrained. Although it has been assigned to the Early Devonian (Zanettini, 1981), Manceñido and Damborenea have argued that the paleontologic record is too poor to accurately define the age of the Herrada Member and that it could also be Late Silurian. The upper member, therefore is here conservatively considered as Late Silurian–Early Devonian. According to Cortés *et al.* (1984), the fossil assemblage found in the lower member of the Sierra Grande Formation indicates warm waters.

In the Sierra Grande area, outcrops of this formation can be divided into three main exposures: *(i)* northern (Fig. 1A), *(ii)* southern; and *(iii)* eastern. The ferriferous levels are present only in the first two exposures. Paleomagnetic sampling was carried out only in the northern outcrop, because an important hydrothermal metamorphism, due to a late Paleozoic granitic intrusion, has widely affected the mineralogy of the ferriferous horizons in the southern exposure (Gelós, 1977; Zanettini, 1981). On the other hand, Avila (1980) and Zanettini (1981) have stated that "the original mineralogical characteristics of the depositional processes seem to be preserved in the ferriferous beds of the northern exposure." These are composed mainly of oolitic hematite, with minor proportions of chlorite and magnetite. The genesis of these ferriferous horizons has been widely discussed in Argentine literature (Valvano, 1954; Avila, 1980; Zanettini, 1981; Cortés *et al.*, 1984), it being generally accepted that in the northern exposure, the oxides (hematite and magnetite) are the original chemical precipitates or early post–depositional products of the dehydration of amorphous ferric and ferrous hydroxides.

The Sierra Grande Formation is extensively deformed in N– to NNW–trending folds with associated faults (Fig. 1B). In the northern exposure, the rocks are folded in three parallel north–south folds with a wave length of about 5 km. The sampling area (Fig. 1A) is located on the western flank of the Rosales Syncline (Zanettini, 1981). After the folding, several faults disrupted the sequence, causing local variations in the attitudes of the beds (see Tables 1–3). The most important tectonic phase took place after deposition of these sedimentary rocks and prior to the development of the mid–Carboniferous (330 Ma) magmatic complex of the North Patagonian Massif (Llambias *et al.*, 1984). The approximate age of this tectonic phase is generally considered to be Middle Devonian to Early Carboniferous. Ramos (1984) and Ramos and Cortés (1984) related these movements to the Late Devonian Chañic event (Aceñolaza and Tosselli, 1973) and proposed that it was the result of the beginning of the collision of Patagonia with the Brazilian Craton. After this tectonic phase, widely distributed intrusions of granitic rocks ocurred in many localities of Patagonia. In the Sierra Grande area, they are mainly of Early Permian age (Rb/Sr 261±5 Ma; Halpern, 1972, in Cortés *et al.*, 1984) and have been correlated with the Sierra Paileman Formation (Stipanicic and Methol, 1972), dated as 270±10 Ma (Halpern *et al.*, 1970, in Llambias *et al.*, 1984). Thick rhyolitic flows, the Marifil Formation, erupted during the Early to Middle Jurassic. Some co–genetic dikes intruded the Sierra Grande Formation in the sampling area (Fig 1)

PALEOMAGNETIC STUDY

Sixty–eight hand samples, oriented with sun and magnetic compass, were collected from the Sierra Grande Formation. Another twelve samples were obtained from two rhyolitic dikes of the Marifil Formation. Locations of sampling sites are shown in Fig. 1A. Because the hand samples of the Sierra Grande Formation were collected from different stratigraphic levels, only a within–sample (not within–site) consistency of the data could be attempted — the only exception being the rhyolitic dikes. Sites A, F, and G consist of long–exposed sections from which samples were systematically collected at different levels (similar to magnetostratigraphic sampling). The sections were subdivided according to variations in attitude of the beds along them (*e.g.*, F1, F2, etc.).

Rosales Ferriferous Horizon

Twenty hand samples were collected from the Rosales horizon, which is situated in the lower member of the formation. Ten samples (SG–212 to –221) were systematically collected from different stratigraphic levels at site B (Fig. 1A). Three others belonged to site C (SG–222 to –224). Only two samples each were obtained at sites D (SG–231 and –232) and E (SG–235 and –236). Finally, three

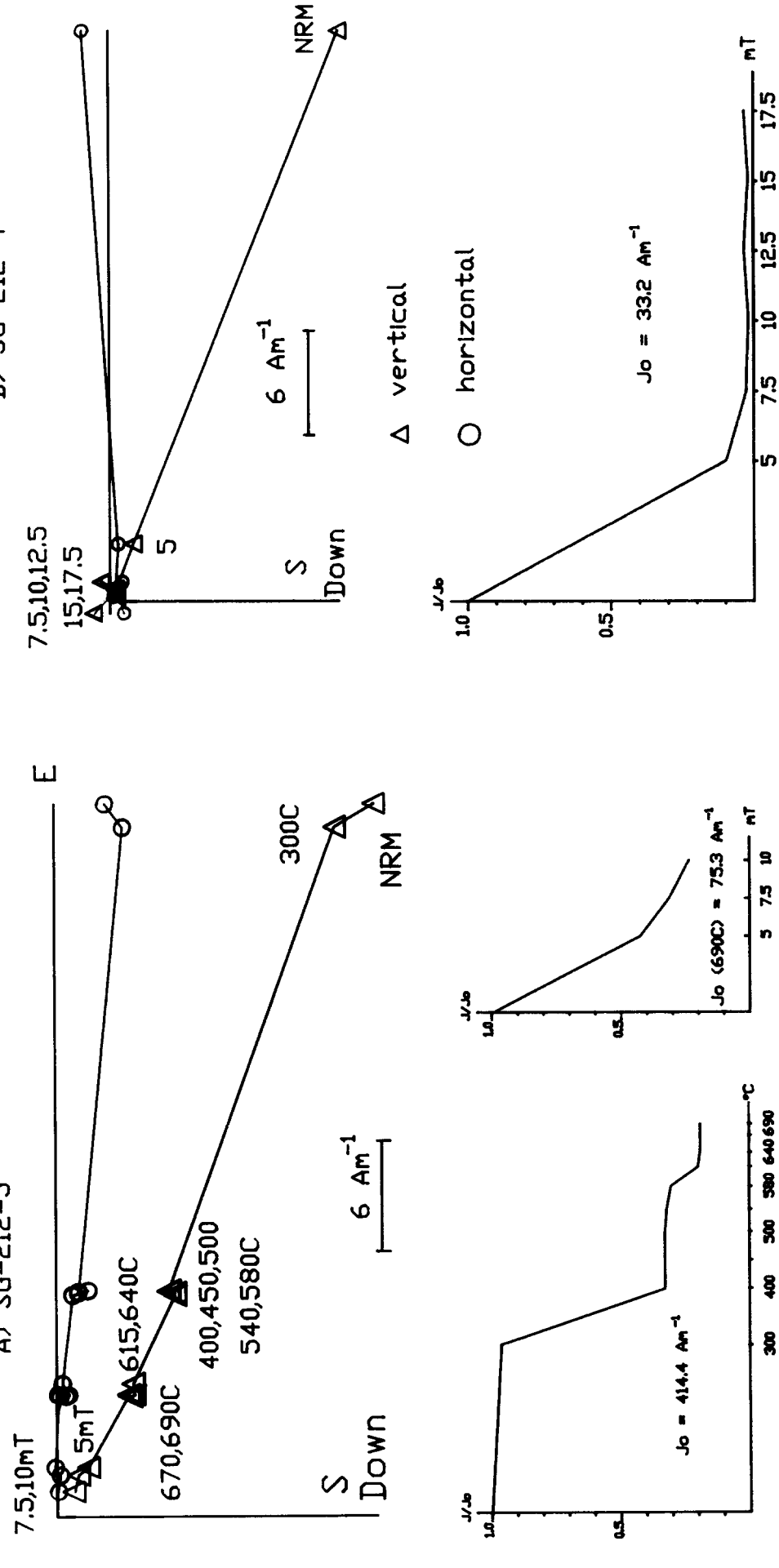


Fig. 2. Magnetic behavior of two specimens of a sample of the Rosales ferriferous horizon that underwent thermal (A) and AF (B) cleaning. Sample SG-212-5 underwent stepwise AF demagnetization up to 10 mT after thermal cleaning at 690°C. Magnetic behavior is represented by orthogonal vector diagrams and normalized demagnetization curves.

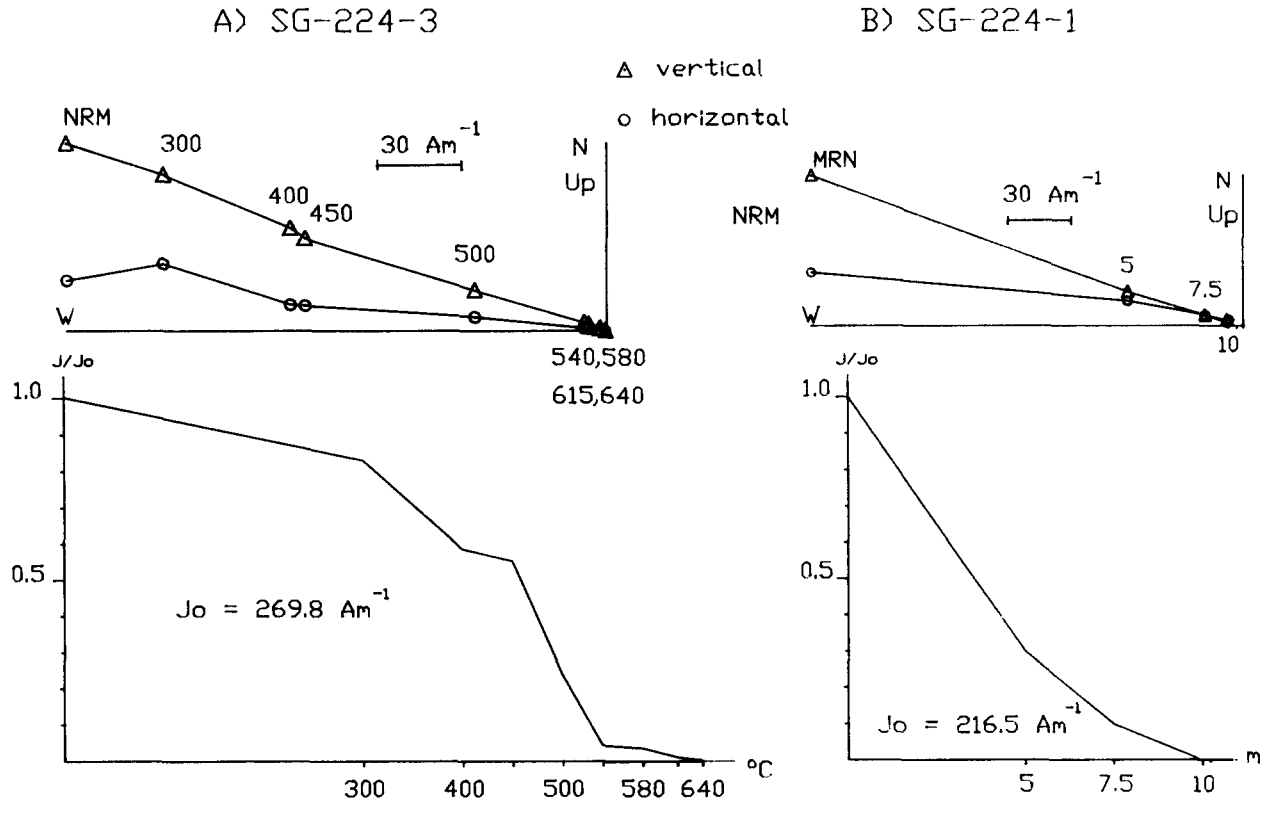


Fig. 3. Magnetic behavior of two specimens of a sample of the Rosales ferriferous that underwent thermal (A) and AF (B) cleaning, represented by orthogonal vector diagrams and normalized demagnetization curves.

samples (SG-277 to -279) were collected from the ferriferous horizon at site F2.

Two to five standard specimens were processed from each sample. All showed extremely high intensities of NRM (10^2 Am $^{-1}$). Figures 2 and 3 show the characteristic magnetic behavior of the samples from the ferriferous horizon. Most of the samples showed univectorial magnetizations, with the exception of a superimposed, easily cleaned, viscous magnetization that was present in a few samples. Fig. 2 shows the magnetic characteristics of two specimens from SG-212 that were subjected to thermal (Fig. 2A) and alternating field (AF, Fig. 2B) demagnetizations. Both techniques demonstrate the existence of a univectorial magnetization that is very stable at very high temperatures (690°C) but very soft when subjected to AF cleaning. The high thermal stability and low coercivities of the magnetization carried by these rocks are not fully understood but, according to their mineral compositions and demagnetization curves, it seems likely that it is carried by magnetite (titanomagnetite?) and hematite. The hematite also showed low coercivities, as shown by AF cleaning applied to SG-212-5, previously demagnetized up to 690°C. Specimens presented in Fig. 3A and 3B show similar characteristics but antipodal directions. The magnetization also seems to be carried by both magnetic minerals, although the amount of hematite seems to be much less than in the former (only 5% of the NRM remains after 580°C). According to the characteristics shown by these samples, the stable remanent magnetization (SRM) was isolated by the average of the directions of residual remanent magnetizations at three or more different stages. The directions isolated for the specimens of each sample were averaged in order to get one SRM direction for each sample and to assess the within-sample consistency of the results. All of these parameters are given in Table 1.

Figure 4 shows the SRM directions isolated for each sample of the Rosales horizon before (Fig. 4A) and after (Fig. 4B) structural corrections are applied. The structural corrections were performed by rotating the beds around horizontal axes coincident with strike. This simple tilt correction could have produced apparent rotations (MacDonald, 1980) in the remanence directions. Nevertheless, the fact that no large systematic differences in declination are apparent among the sites suggests that if any apparent rotation was produced, it was small. In Fig. 4, the triangles represent those directions (marked with an asterisk in Table 1) considered to be recordings of a transitional field. The recording of a transition of polarity is likely preserved at site B, where samples SG-212 to SG-221 were systematically collected from bottom to top of the ferriferous horizon. Samples SG-215, -216 and -217 represent this transition of polarity recorded along a section 1.5 meters thick. The transition is shown in the stereo-net of Fig. 4B. The other two anomalous directions (SG-231 and SG-232), both from site D, may also represent a transitional field or other anomalies associated with this site. All these anomalous directions were not taken into account when computing the mean direction of the remanence of the Rosales horizon. As the struc-

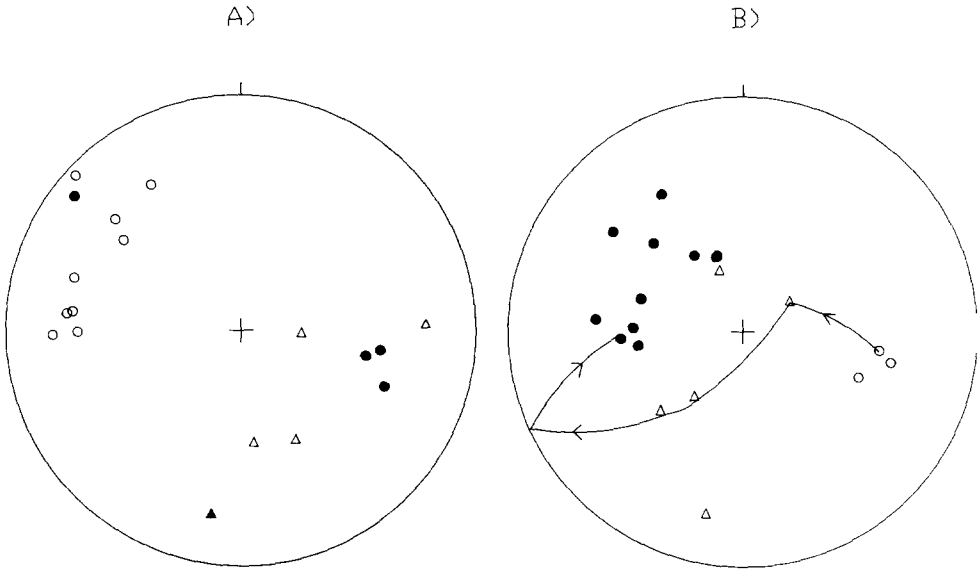


Fig. 4. **A)** *In situ* stable remanent magnetization (SRM) directions isolated for each sample of the Rosales horizon (see Table 1). **B)** Idem after the application of the structural corrections. Each sample belongs to a different stratigraphic level. The arrows indicate the suggested transition of polarity recorded at site B. Closed (open) symbols indicate positive (negative) inclinations. The triangles indicate SRM directions representative of transitional fields. References in the text.

Table 1. Paleomagnetic data on the Rosales horizon (Sierra Grande Formation).

			SRM				Attitude		Corrected SRM				Pole Position			
Site	Sample	N	Dec (deg)	Inc (deg)	K	α_{95} (deg)	Strike (deg)	Dip (deg)	Dec (deg)	Inc (deg)	K	α_{95} (deg)	Lat (deg)	Long (deg)	D1 (deg)	D2 (deg)
B	SG-212	4	101.1	33.2	20	21.2	17	58 SE	101.6	−24.5						
	SG-213	2	111.0	23.8	262	15.5	17	58 SE	111.4	−34.0						
	SG-214	2	97.8	28.2	803	8.8	17	58 SE	97.7	−29.1						
	SG-215*	3	87.2	−13.5	104	12.2	17	58 SE	58.1	−64.2						
	SG-216*	2	153.5	−34.7	43	39.1	17	58 SE	214.1	−51.4						
	SG-217*	3	173.0	−39.1	87	13.3	17	58 SE	224.5	−37.0						
	SG-218	3	269.3	−20.3	410	6.1	17	58 SE	266.6	35.0						
	SG-219	2	311.2	−19.3	12	78.9	17	58 SE	314.7	33.7						
	SG-221	1	312.9	−2.3	—	—	17	58 SE	327.5	47.8						
C	SG-222	3	287.4	−17.0	60	16.0	17	58 SE	287.5	41.0						
	SG-223	2	268.4	−12.6	116	23.4	17	58 SE	262.3	42.0						
	SG-224	2	275.3	−16.8	299	14.5	17	58 SE	272.4	39.9						
D	SG-231*	3	93.3	−61.3	33	21.7	39	52 SE	340.9	−57.9						
	SG-232*	2	190.8	12.2	1791	5.9	39	52 SE	191.3	−13.5						
E	SG-235	3	307.8	6.3	184	9.1	4	65 SE	339.7	52.7						
	SG-236	2	308.0	6.7	2057	5.5	4	65 SE	340.4	52.7						
F2	SG-277	2	328.1	−18.1	295	14.6	26	47 SE	329.1	22.1						
	SG-278	2	307.3	−25.7	116	23.3	26	47 SE	306.9	20.5						
	SG-279	3	276.2	−18.5	302	7.1	26	47 SE	275.1	25.9						
Mean		14	293.2	−16.8	14	11.0			296.1	38.6	12	12.0	3.4	238.0	14.3	8.5
									(rotated to African coordinates)				−34.2	283.7	14.3	8.5
									(without structural correction)				23.0	214.0	11.4	5.9
									(rotated to African coordinates)				−17.1	255.5	11.4	5.9

Key: SRM, stable remanent magnetization; K, statistical precision parameter; α_{95}, radius of the 95% confidence circle (Fisher, 1953); D1 and D2, long and short semi-axes of the 95% confidence oval; *SRM directions representative of transitional fields, not considered in the means; north latitudes and east longitudes, positive. Rotation to African coordinates according to reconstruction of Smith and Hallam (1970); more references in text.

Table 2. Paleomagnetic data on the San Carlos Member (not considering the Rosales horizon).

			SRM				Attitude		Corrected SRM				Pole Position*			
Site	Sample	N	Dec (deg)	Inc (deg)	K	α_{95} (deg)	Strike (deg)	Dip (deg)	Dec (deg)	Inc (deg)	K	α_{95} (deg)	Lat (deg)	Long (deg)	D1 (deg)	D2 (deg)
A1	SG-203		286.8	51.2			359	62 E	64.0	63.2						
	SG-204		250.2	38.2			359	62 E	141.2	71.3						
A2	SG-207		245.1	52.2			10	51 E	149.7	62.6						
	SG-208		358.2	72.4			10	51 E	77.2	40.4						
	SG-211		343.1	68.5			10	51 E	72.2	45.6						
Mean sites A		5	278.7	63.4	9	27.1			91.7	61.2	12	23.0				
F1	SG-262		202.2	75.9			48	52 SE	152.8	30.9						
	SG-264		237.3	72.1			48	52 SE	160.9	38.7						
	SG-265		109.5	68.4			48	52 SE	127.3	18.5						
	SG-266		243.4	70.3			48	52 SE	163.3	40.6						
	SG-267		207.5	73.1			48	52 SE	156.4	30.6						
	SG-268		244.4	72.7			48	52 SE	160.1	40.8						
	SG-270		213.6	68.8			48	52 SE	161.9	30.2						
F2	SG-272		246.8	49.9			26	47 SE	176.9	56.1						
	SG-273		195.7	52.8			26	47 SE	158.2	27.7						
	SG-274		55.2	74.3			26	47 SE	99.4	34.1						
	SG-276		38.3	−67.3			26	47 SE	327.4	−43.6						
Mean sites F		11	215.7	74.1	19	10.6			150.8	37.2	18	11.1				
Mean sites A, F		16	241.2	73.4	12	11.0			139.2	47.1	8	13.6	−49.0	251.5	19.7	17.7

Key: SRM, stable remanent magnetization; K, statistical precision parameter; α_{95}, radius of the 95% confidence circle (Fisher, 1953); D1 and D2, long and short semi-axes of the 95% confidence oval. *Poles computed without structural corrections.

tural corrections corresponding to the different sites were not significantly different (Graham, 1949; McFadden and Jones, 1981), a tilt test could not be applied. Nevertheless, the SRM directions passed a reversal test, antipodal directions, at the 95% confidence level (McFadden and Lowes, 1981). The following factors permit acceptance of the SRM isolated as an early chemical remanent magnetization (CRM): the univectorial character of the magnetization; the well defined demagnetization end-points in most of the samples; the high stability of the magnetization up to 690°C; the same direction apparently carried by magnetite and hematite, which are considered to be early (perhaps primary) minerals; an acceptable within-sample consistency of the data; a positive reversal test; and the probable recording of a reversal of the magnetic field. The structural corrected mean direction was used to compute a PP (SG1) of the Rosales horizon: 238.0°E, 3.4°N, N=14, D1=14.3°, D2=8.5°; where D1 and D2 are the long and short semi-axes of the 95% confidence oval, repectively. The fact that the position of the pole computed from the uncorrected SRM (214.0°E, 23.0°N or 255.5°E, 17.1°S if rotated to Gondwana framework; Smith and Hallam, 1970) does not coincide with any post-Silurian section of the Gondwana apparent polar wander (APW) path (*e.g.*, Vilas, 1981) can be taken as additional evidence for a pre-tectonic magnetization. According to the criteria recently established by Van der Voo (1988), the quality factor of this PP is estimated at 5 (maximum 7), because the criteria of at least 25 samples (although more than 30 specimens were considered here) and a positive field test (the recording of a polarity transition is not taken here as such a test) could not be accomplished.

The San Carlos Member (Lower Member)

Twenty-three hand samples were collected from rocks of The San Carlos Member (not counting those from the Rosales horizon). Three were taken at site A1 (SG-203 to -205); six at site A2 (SG-206 to -211); nine (originally ten but one was lost) at site F1 (SG-262 to -270); and five at site F2 (SG-272 to -276). All of the samples were fine sandstones and, except those from site F1, belong to stratigraphic levels younger than the Rosales horizon. Paleomagnetic data obtained from these samples are presented in Table 2.

Figure 5 shows the typical magnetic behavior of samples from site F (F1 and F2). The AF technique was not useful for destroying magnetization (Fig. 5A), whereas thermal demagnetization showed a univectorial magnetization stable at high temperatures (630°C), probably carried by hematite. Some samples (Fig. 5B) also showed a superimposed magnetic component, probably viscous, that was easily removed. In comparison, samples from site A (A1 and A2) presented a very soft magnetization, easily removed by AF cleaning (Fig. 6). SRM could be isolated in only five samples from site A, but was isolated in eleven

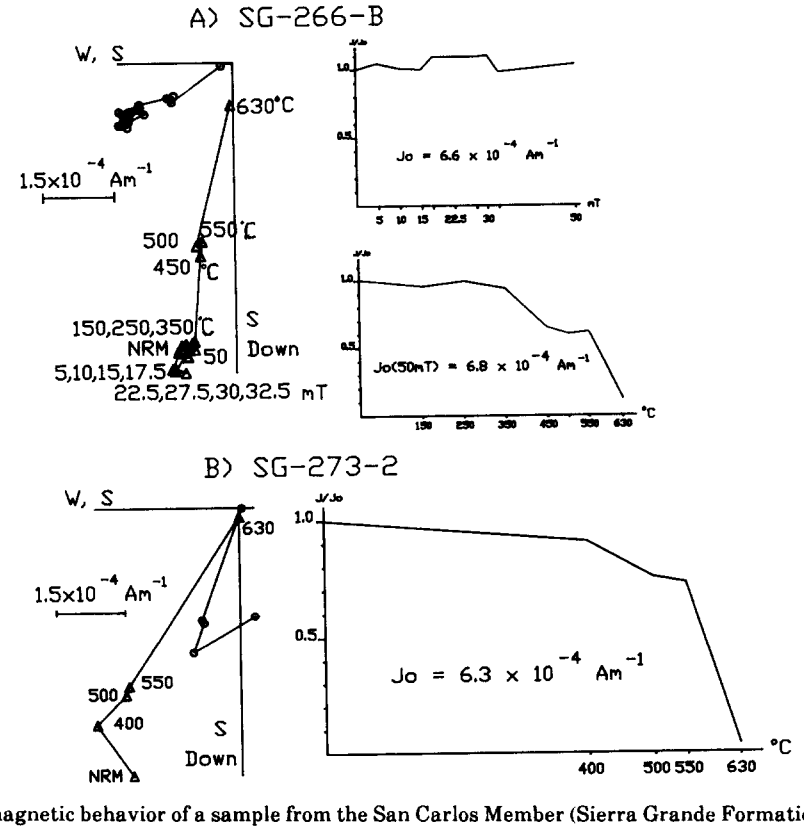


Fig. 5. A) Typical magnetic behavior of a sample from the San Carlos Member (Sierra Grande Formation), collected at site F1 and submitted to combined AF-thermal demagnetization. B) Magnetic behavior of another sample, collected at site F2 and submitted to thermal cleaning. References in the text.

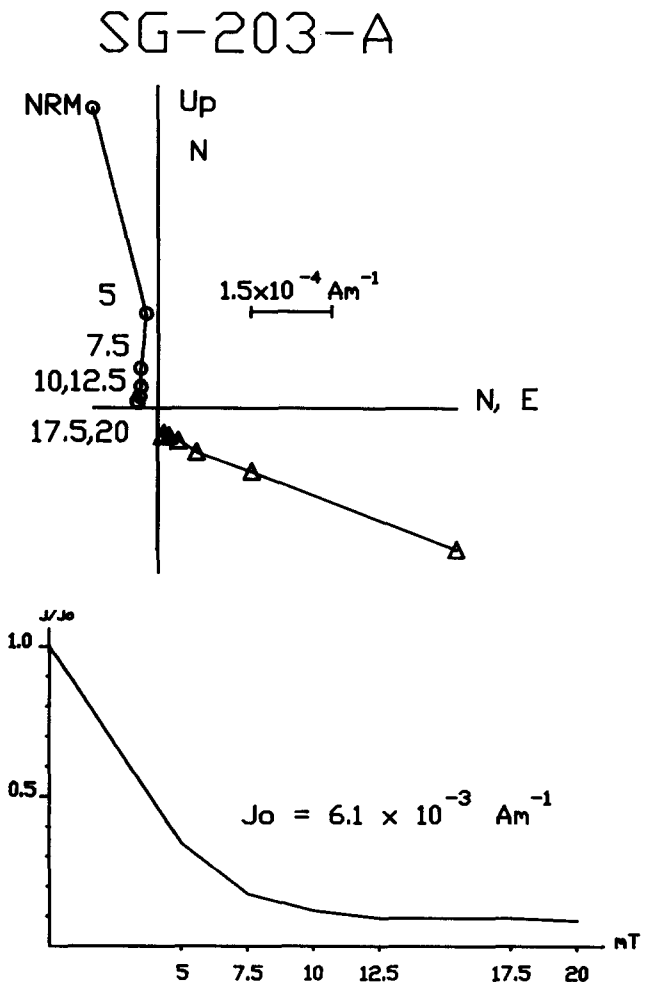


Fig. 6. Magnetic behavior of a sample of the San Carlos Member collected at site A. An example of the soft magnetization carried by these samples.

from site F (Table 2). The SRM directions isolated from each sample before and after applying the structural corrections are shown in Fig. 7A and 7B, respectively. The directions from site A are represented by triangles, those from site F by circles. The 95% confidence circle of the mean directions is also shown. The better grouping of the directions before applying structural corrections is evident, consistent with the improvement in statistical parameters (see Table 2), which would suggest post-tectonic secondary magnetization of the fine sandstones of the San Carlos Member. The same happens if only the directions from site F are considered. Despite that, the result of the tilt test is undefined, as SRM populations from sites A and F are distinguished statistically at a 95% confidence limit before and after applying the corrections. This, together with the different magnetic properties, suggests that the magnetizations isolated at sites A and F may represent different magnetic processes. Nonetheless, they will be considered, conservatively, as post-tectonic secondary magnetizations presumably carried by hematite — at least at site F. A pole position was then computed for site F and from all the samples on the basis of non-corrected SRM directions. These are shown in Table 2, together with their statistical parameters. When only site F is considered, the pole position obtained is consistent with the Permian-Early Jurassic poles of Gondwana (Oviedo and Vilas, 1984), suggesting a remagnetization related to the Permian granitic intrusions.

The Herrada Member (Upper Member)

Twenty-four hand samples, sixteen at site G1 (SG-280 to -295) and eight at site G2 (SG-296 to -303), were collected from red and green siltstones and fine sandstones of the upper member of the Sierra

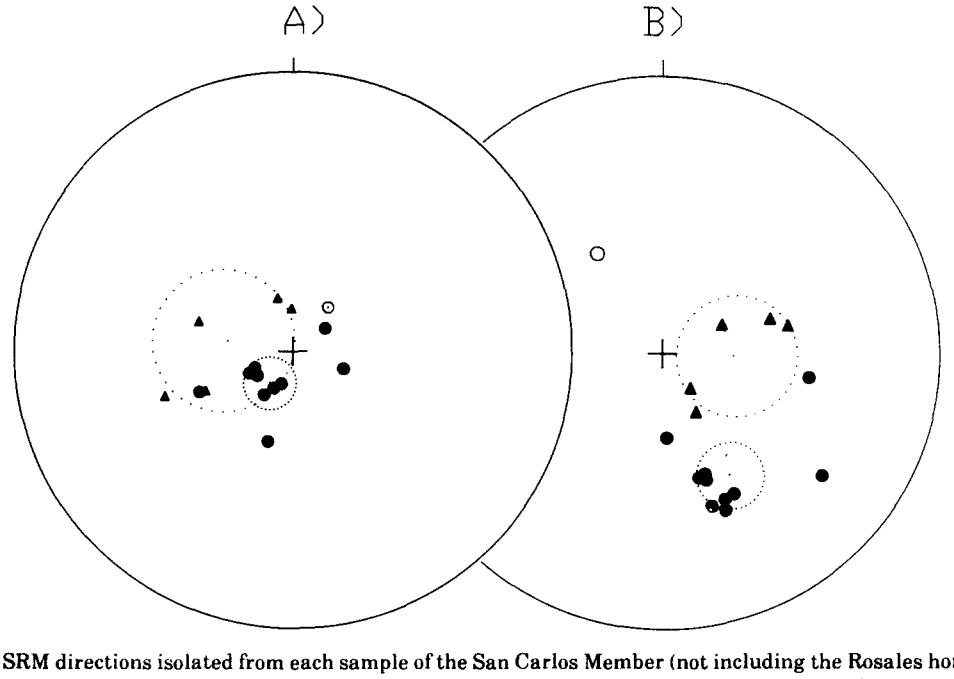


Fig. 7. A) *In situ* SRM directions isolated from each sample of the San Carlos Member (not including the Rosales horizon). Circles, samples from site F; triangles, samples from site A. The major circles represent 95% confidence circles of their respective means. Closed (open) symbols indicate positive (negative) inclinations. B) Idem after the application of the structural corrections.

Grande Formation. At least two specimens from each sample were processed. The specimens were subjected to thermal demagnetization or chemical leaching by hydrochloric acid (7N). Pilot treatments by AF proved to be useless. The magnetic behavior of these samples is shown in Figs. 8 and 9. Two different superimposed magnetic components were isolated, mainly by thermal demagnetization (Fig. 8A), although sometimes chemical leaching was also useful. These magnetic components are a northwesterly component of negative inclination, generally removed at about 500°C, and a steep, northwesterly, high-temperature component of positive inclination. Although most of the samples carried both magnetic components, these could not always be isolated by demagnetization treatments (Fig. 8B). Few samples showed only one magnetic component (Fig. 9). The low-temperature negative inclination component was isolated in eleven samples, being considered a secondary magnetization. These directions, as well as the 95% confidence circle of the mean, are shown in Fig. 10. The high temperature and positive component were considered to be isolated only in those cases in which undoubted demagnetization end-points could be reached (*e.g.*, Fig. 8A). This component could be isolated in only ten samples, five from each site (Table 3). An objective within-sample consistency of isolated directions could not be determined in many cases, but in all, more than one specimen from each sample showed the existence of the positive inclination component, although demagnetization end-points could not be reached.

The SRM directions (the high temperature positive inclination component) of this member are shown before and after correction for structural attitude (Fig. 11A and 11B, respectively). The directions from site G1 are represented by circles, those from site G2 by triangles. The 95% confidence circles of the respective means are also shown. The better grouping of the directions after correction is evident, and the improvement in statistical parameters is shown in Table 3. This indicates a pre-tectonic magnetization which was confirmed by application of the fold test (McFadden and Jones, 1981).

Considering that the main tectonic event affecting the Sierra Grande Formation is probably Late Devonian in age (Ramos and Cortés, 1984), this magnetic component is considered to be a primary magne-

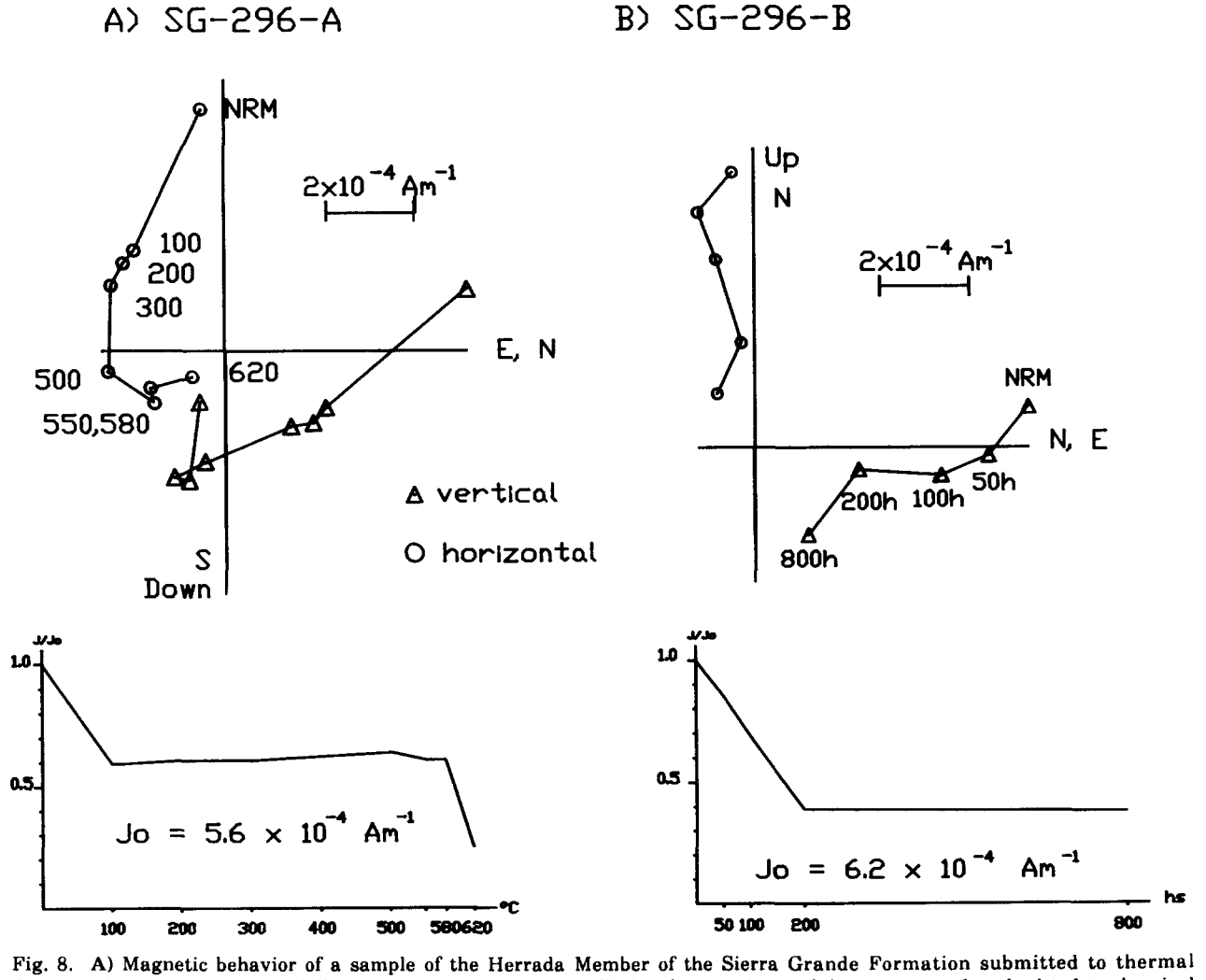


Fig. 8. A) Magnetic behavior of a sample of the Herrada Member of the Sierra Grande Formation submitted to thermal demagnetization showing two different magnetic components. B) Another specimen of the same sample submitted to chemical leaching without positive results. References in the text.

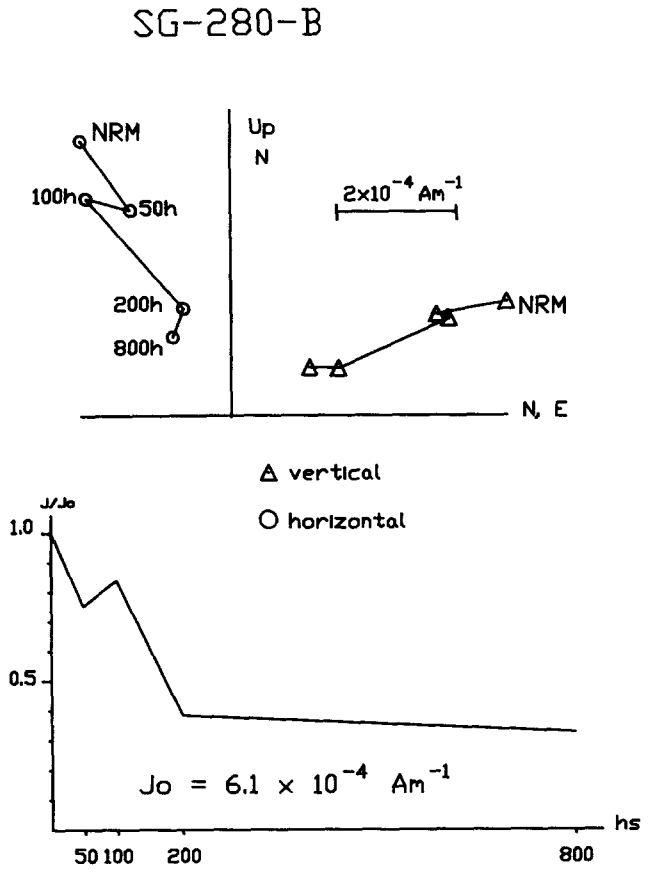


Fig. 9. Magnetic behavior of a sample of the Herrada Member submitted to chemical leaching, showing only one magnetic component. References in the text.

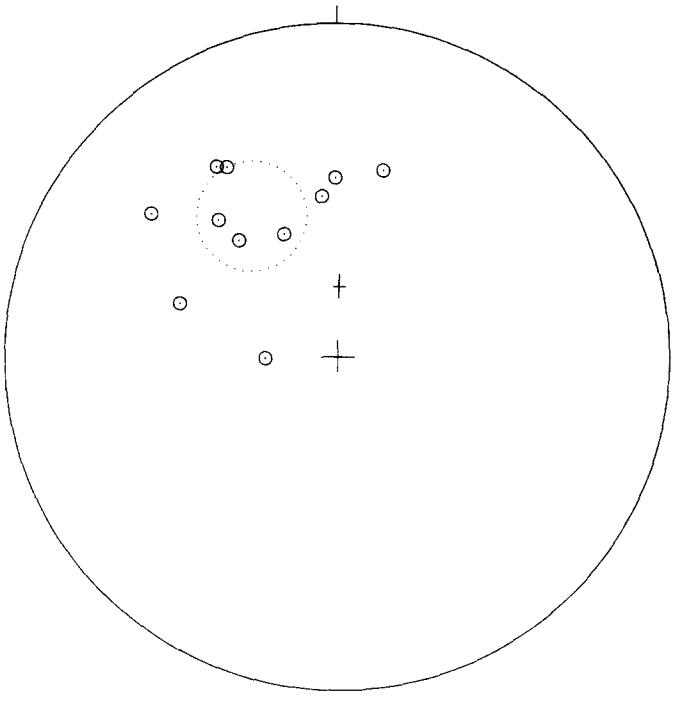


Fig. 10. *In situ* directions of the low-temperature component isolated in samples of the Herrada Member and the 95% confidence circle of the mean. The small cross represents the present dipole. Open symbols indicate negative inclination.

tization. Accordingly, a PP (SG-2) was computed on the basis of the pre-tectonic mean direction of magnetization: 283.5°E, 42.0°S, N=10, D1=20.4°, D2=20.2°. Using Van der Voo's criteria mentioned above, this pole has a quality of 4 or 5, its weakest points being the small number of samples and the lack of antipodal reversals. However, according to the most recently published chronostratigraphic time scale (IUGS, 1989), the Early Devonian would have been a period of dominantly reversed polarity.

Jurassic Rhyolitic Dikes (Marifil Formation)

Twelve hand samples, six from each dike, were collected at sites H (SG-225 to -230) and I (SG-256 to -261). Most of the samples showed a univectorial

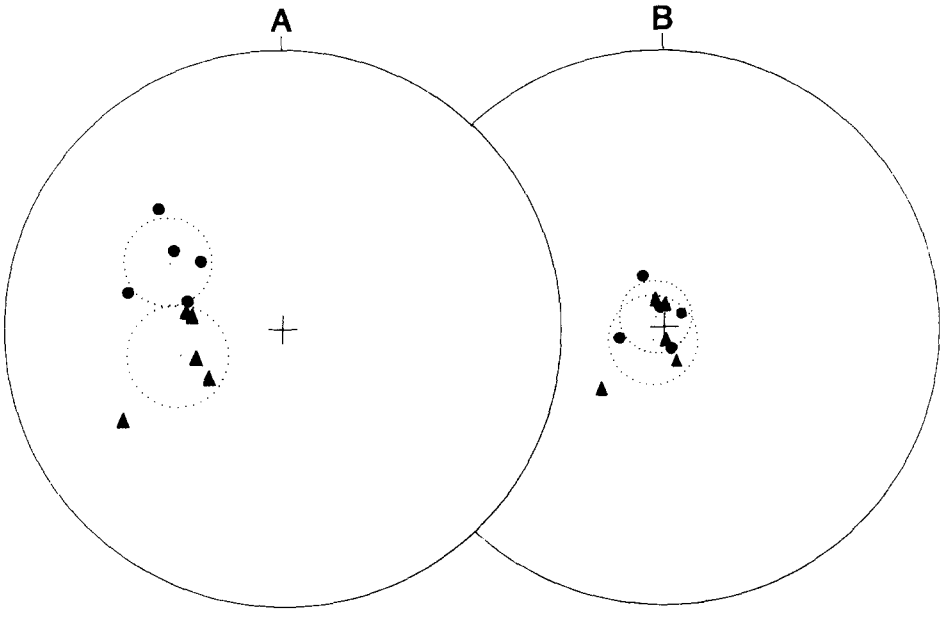


Fig. 11. Positive fold test of the high-temperature component directions isolated in samples of the Herrada Member. A) *In situ* directions and the 95% confidence circles of the means. Circles, samples from site G1; triangles, samples from site G2; closed symbols, positive inclination. B) Idem after the structural corrections.

Table 3. Paleomagnetic data on the Herrada Member.

			SRM				Attitude		Corrected SRM				Pole Position*			
Site	Sample	N	Dec (deg)	Inc (deg)	K	α_{95} (deg)	Strike (deg)	Dip (deg)	Dec (deg)	Inc (deg)	K	α_{95} (deg)	Lat (deg)	Long (deg)	D1 (deg)	D2 (deg)
G1	SG-280		287.4	51.5			28	45 SE	161.6	80.4						
	SG-285		314.3	26.4			28	45 SE	338.6	67.2						
	SG-290		283.2	30.6			28	45 SE	254.3	71.5						
	SG-291		306.4	39.0			28	45 SE	347.0	81.4						
	SG-293		310.0	48.1			28	45 SE	53.0	81.2						
Mean site G1		5	300.6	39.8	28	14.8	28	45 SE	319.3	84.5	28	14.8				
G2	SG-296		236.5	54.3			353	36 E	161.0	74.6						
	SG-297		278.4	53.2			353	36 E	354.3	80.9						
	SG-299		252.5	53.4			353	36 E	174.6	83.7						
	SG-300		241.1	23.4			353	36 E	226.1	55.3						
	SG-301		278.9	50.9			353	36 E	341.7	79.9						
Mean site G2		5	256.1	48.6	19	18.1	353	36 E	221.8	83.1	19	18.1				
Mean both sites		10	280.3	46.4	12	14.5			263.8	85.9	23	10.3	−42.0	283.5	20.4	20.2
									(rotated to African coordinates)				−45.5	6.0	20.4	20.2

Key: SRM, stable remanent magnetization; K, statistical precision parameter; α_{95}, radius of the 95% confidence circle (Fisher, 1953); D1 and D2, long and short semi-axes of the 95% confidence oval. *Poles computed with structural corrections. Rotation to African coordinates according to reconstruction of Smith and Hallam (1970); more references in text.

magnetization with a northerly negative inclination, easily isolated by AF treatment (Fig. 12A). Others showed instead a high-intensity northwesterly positive component superimposed to the former, but which was easily removed by AF of 10 to 15 mT (Fig. 12B). As the samples correspond to dikes, no structural corrections could be applied. The isolated SRM directions are shown in Fig. 13A. A within-site consistency of the directions could be determined considering each dike as an individual cooling unit. Two virtual geomagnetic poles (VGP), one from each dike, were computed on the basis of the site mean directions (Table 4); the average of both is shown in Fig. 13B, together with its 95% confidence oval. All these paleomagnetic data are presented in Table 4. The mean geomagnetic pole position is consistent with that obtained by Mena (in press), shown as a square, from several ignimbritic flows of the Marifil Formation exposed 100 km north of Sierra Grande town. This suggests that the magnetization of the rhyolitic dikes is primary, although no field test could be done. The completely different magnetizations of these rocks with respect to those of the Sierra Grande Formation also suggest that the Jurassic volcanism did

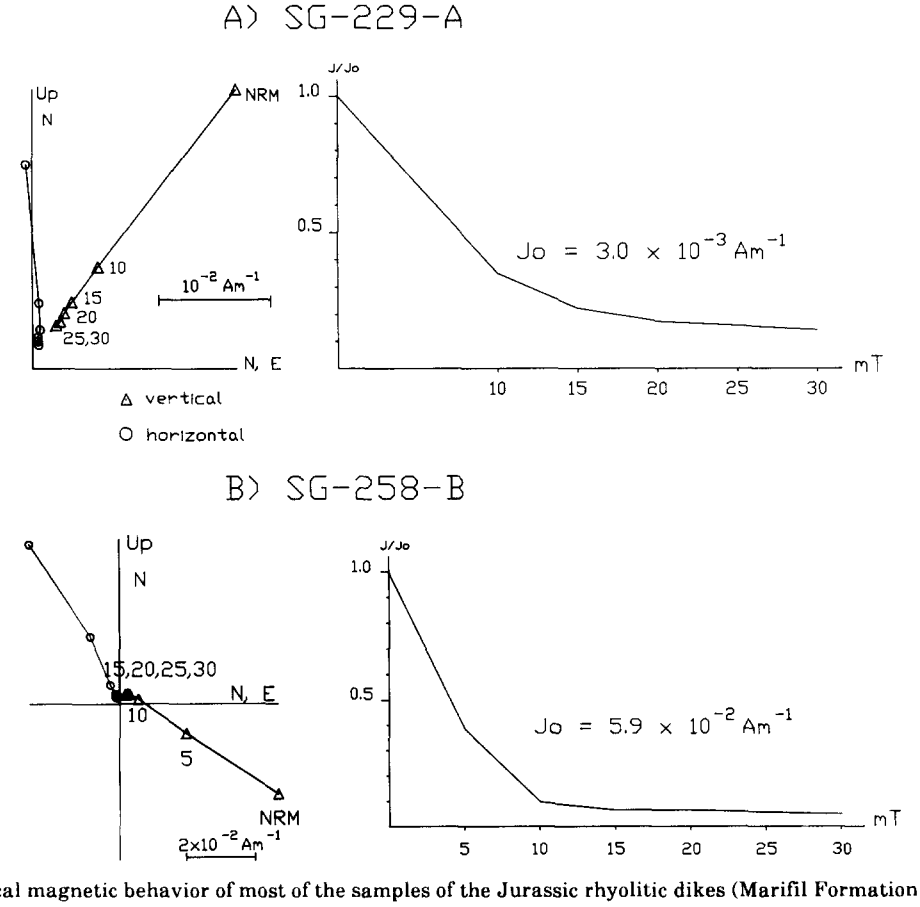


Fig. 12. A) Typical magnetic behavior of most of the samples of the Jurassic rhyolitic dikes (Marifil Formation) submitted to AF demagnetization. B) Example of a sample with an easily removed superimposed high-intensity magnetic component with opposite inclination.

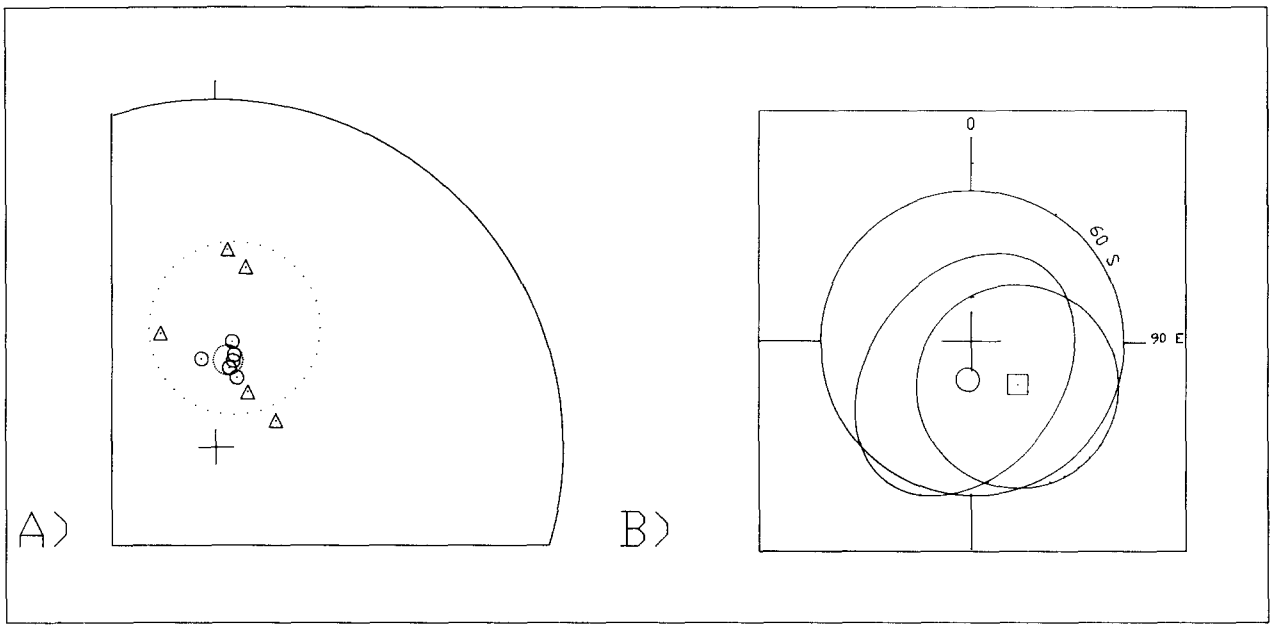


Fig. 13. A) SRM directions isolated from each sample of the Jurassic dikes. Small circles, samples from site H; triangles, samples from site I. The 95% confidence circle of each site mean is also shown. Open symbols, negative inclinations. B) Coincidence of the mean geomagnetic pole of both Jurassic dikes (small circle) and its 95% confidence oval with that obtained previously for the Marifil Formation in other areas (square, Mena, in press) and its respective a_{95}.

not affect the magnetization of the middle Paleozoic rocks in this area.

INTERPRETATION AND DISCUSSION

The paleomagnetic poles obtained from the Rosales horizon (SG1) and from the upper member of the Sierra Grande Formation (SG2) are the first middle Paleozoic poles from Patagonia. Although they have to be considered preliminary poles, in view of the small number of samples, they are considered reliable enough to be used in tectonic and paleogeographic interpretations. Because of the paleontologic assemblages found in the lower member of the formation (Manceñido and Damborenea, 1984), the age of SG1 seems to be well constrained to the Wenlockian (428–424Ma; I.U.G.S., 1989). However, the age of SG2 is not so well constrained and is assigned to the Late Silurian–Early Devonian.

To use these PPs in order to elucidate the tectonic history of Patagonia with respect to the Brazilian Craton, it is necessary to compare these poles with those from the Gondwanic continents (Van Houten and Hargraves, 1987). This is not simple, because the middle Paleozoic apparent polar wandering path (a.p.w.p.) of Gondwana is still a matter of controversy. In the last two decades, three substantially different early–middle Paleozoic Gondwana paths have been proposed (McElhinny and Embleton, 1974; Schmidt and Morris, 1977; Morel and Irving, 1978). These three main paths are represented in Fig. 14, slightly modified according to recent data. For clearness, only the most reliable middle Paleozoic poles from Gondwana (Li *et al.*, 1988; Schmidt and Li, 1989) are represented in Fig. 14, although the paths

Table 4. Paleomagnetic data on the Jurassic rhyolitic dikes.

			SRM				Pole Position*			
Site	Sample	N	Dec (deg)	Inc (deg)	K	a_{95} (deg)	Lat (deg)	Long (deg)	D1 (deg)	D2 (deg)
H	SG-225		16.8	−66.4						
	SG-226		11.3	−61.7						
	SG-227		350.4	−61.2						
	SG-228		9.0	−55.9						
	SG-229		11.2	−59.9						
	SG-230		9.6	−64.1						
Mean site H		6	7.9	−61.8	212	4.6	−84.0	220.4	7.1	5.5
I	SG-257		9.7	−34.8						
	SG-258		334.0	−50.0						
	SG-259		3.5	−30.9						
	SG-260		30.1	−69.2						
	SG-261		66.3	−68.6						
Mean site I		5	8.9	−53.7	10	25.0	−79.8	161.0	34.9	24.4
Mean both sites		2	8.4	−57.7	201	17.7	−82.8	180.9	26.0	19.1

Key: SRM, stable remanent magnetization; K, statistical precision parameter; a_{95}, radius of the 95% confidence circle (Fisher, 1953); D1 and D2, long and short semi-axes of the 95% confidence oval. *Poles computed without structural corrections.

are based upon many more paleomagnetic poles. A nearly complete list of these poles can be found in Van Houten and Hargraves (1987). The early Paleozoic poles from the Gondwanic continents lie, in general, when rotated to a Gondwana framework, in northern Africa (Van Houten and Hargraves, 1987), as shown by the location of the mean Middle–Upper Ordovician pole (Van der Voo, 1988). Those of the late Paleozoic are located in southeastern Africa and Antarctica, as shown by the mean Late Carboniferous–Early Permian pole of Gondwana C–P (Oviedo and Vilas, 1984; see also Van Houten and Hargraves, 1987). The major problem in defining the middle Paleozoic section of the Gondwana a.p.w.p. is that most of the Silurian–Devonian paleomagnetic poles have been derived from the Tasman (Lachlan) fold belt in southeastern Australia. Many of these poles lie southwestward of Africa in southern and southwestern South America (Van Houten and Hargraves, 1987).

The earliest proposed a.p.w.p. (McElhinny and Embleton, 1974; path "X" of Morel and Irving, 1978) suggests a slow displacement of the pole from northern to central Africa in the Ordovician to the Devonian (dashed path in Fig. 14). In this hypothesis, the Lachlan fold belt is considered a displaced terrane with respect to eastern Australia; therefore, its paleomagnetic poles are not representative of Gondwana.

The other two hypotheses considered the Tasman poles as representative of Australia and, therefore, of all of Gondwana (see Schmidt and Embleton, 1987, for a critical review of these poles). One, which could be called the "loop" hypothesis (path "Y," Morel and Irving, 1978; Goleby, 1980), proposes that the Gondwana pole position moves quickly from northern Africa in the Ordovician to southwestern South America

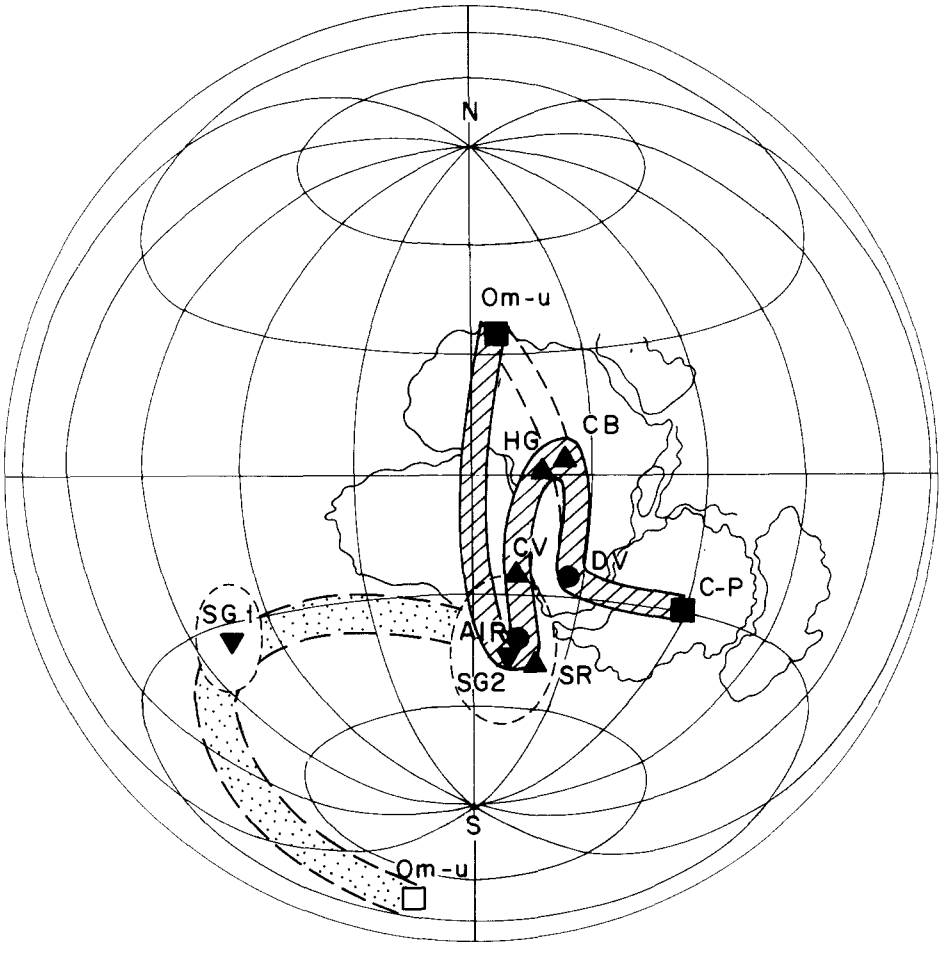


Fig. 14. Three substantially different early to mid-Paleozoic APW paths of Gondwana, slightly modified according to recent data. The dashed path (McElhinny and Embleton, 1974; path "X" of Morel and Irving, 1978) does not consider the Tasman poles as representative of Gondwana. The striped path (path "Y" of Morel and Irving, 1978; Goleby, 1980) or simply the "loop" path, seems to be the most accepted path nowadays, according to the recently obtained African AIR pole. The dotted path considers a reversed polarity option for the early Paleozoic poles (Schmidt and Morris, 1977). Only those mid-Paleozoic poles of highest quality are represented. Legend: closed squares, Gondwana mean poles; Om–u, Middle–Upper Ordovician (Van der Voo, 1988); C–P, Upper Carboniferous–Lower Permian (Oviedo and Vilas, 1984); open square, position of Om–u considering the reversed polarity option; closed triangles, Australian poles; SR, Snowy River volcanics (Late Silurian–Early Devonian, Schmidt *et al.*, 1987); CV, Comerong volcanics (Middle Devonian, Schmidt *et al.*, 1986); HG, Hervey Group (Late Devonian–Early Carboniferous, Li *et al.*, 1988); CB, Canning Basin limestones (Late Devonian, Hurley and Van der Voo, 1987); closed circle, African pole; AIR, Nigerian Ring Complex (435 Ma, Hargraves *et al.*, 1987); inverted triangles, first poles from Patagonia; SG1, the Rosales horizon, Sierra Grande Formation (Wenlockian, this paper); SG2, the Herrada Member, Sierra Grande Formation (Upper Silurian–Lower Devonian, this paper). These new poles are represented with their respective 95% confidence oval. Gondwana reconstruction of Smith and Hallam (1970). African network. More references in the text.

in the Silurian–Devonian, to come back very rapidly to central Africa in the Late Devonian (striped path, Fig. 14)

The third hypothesis (Schmidt and Morris, 1977) considered a "reversed polarity option" for the pre-Devonian paleopoles. According to this, the South pole was not located in northern Africa, but in the antipodes during the early Paleozoic (open symbol Om–u, Fig. 14). During the Silurian it would have undergone rapid movement through the Pacific to reach southwestern South America in the Silurian–Devonian and central Africa in the Late Devonian (dotted path, Fig. 14).

The hypotheses that considered the Tasman poles as representative of all of Gondwana were strongly supported recently by the acquisition of a high-quality Early Silurian (435 Ma) pole from cratonic Africa (Hargraves *et al.*, 1987). The position of this PP (AIR, Fig. 14) is concordant with those from the Lachlan fold belt (*e.g.*, SR, Schmidt *et al.*, 1987). A pole position on southern South America in the Early Silurian (435 Ma) would imply a rapid shift of the pole during the Late Ordovician–early Early Silurian, probably followed by a "quasi-static" period during most of the Silurian and the Early Devonian, as was proposed by Hargraves *et al.* (1987); note the position of the Early Devonian SR in Fig. 14. Other recent investigations in Australia (Schmidt *et al.*, 1986; Li *et al.*, 1988) have better constrained the Middle–Late Devonian section of the path back from South America to central Africa (PPs CV, HG, and CB).

The "loop" path (path "Y" of Morel and Irving, 1978; Goleby, 1980) has been generally favored in paleogeographic reconstructions over the reversed polarity option path (Schmidt and Morris, 1977), partially because some authors have reported a coincidence of the pole path over Africa and South America with the existence of glacial deposits (Crowell, 1981; Caputo and Crowell, 1985; Van Houten and Hargraves, 1987).

The Wenlockian (SG1) and Late Silurian–Early Devonian (SG2) PPs obtained for Patagonia are shown with their respective 95% confidence ovals in Fig. 14, together with the Gondwana poles. SG1 is located in the Pacific Ocean, far from the most reliable Silurian–Devonian poles of Gondwana. SG2 is coincident with the Late Silurian–Early Devonian poles of the Tasman fold belt (as the Early Devonian Snowy River volcanics pole: SR) in southeastern Australia and with the Early Silurian (435 Ma) AIR pole from Africa. SG1 and SG2 indicate very different paleolatitudinal positions of Patagonia during the Wenlockian (425 Ma) — tropical latitudes — and the Early Devonian (400 Ma) — polar latitudes. This means a rapid displacement of Patagonia during the Late Silurian of about 20 cm/year (minimum). It is remarkable that the tropical latitudes indicated by SG1 are consistent with the temperate- to warm-water faunas found in the lower member of the Sierra Grande Formation (Cortés *et al.*, 1984).

The tectonic implications of the PPs obtained will be considered in regard to the different hypotheses about the origin of the Patagonian Block.

The "Allochthonous" Hypothesis

If we assume, as many paleomagnetists do nowadays, that the "loop" path is the most probable a.p.w.p. of Gondwana, then the African AIR pole would probably represent the Early (perhaps late Early) Silurian pole of Gondwana. Considering the similar position of the Australian Late Silurian–Early Devonian PPs like SR, a nearly "stationary" state of the Gondwana supercontinent from the late Early Silurian to the Early Devonian (Van Houten and Hargraves, 1987) probably ocurred. In this case, the position of SG1 supports an allochthonous origin of Patagonia. Nevertheless, the correspondence of the Early Devonian SG2 with those coetaneous poles from Gondwana suggests that Patagonia underwent rapid displacement during the Late Silurian and was very close to (or part of) Gondwana in the Early Devonian. As there was no tectonic event between the deposition of both members of the Sierra Grande Formation, the collision must have ocurred after the deposition of both members. Thus, during the Late Silurian–Early Devonian, Patagonia should have still been approaching Gondwana, being very close to it, according to the position of SG2 concordant with other Gondwana poles. The important tectonic event that affected the Sierra Grande Formation could represent, then, the beginning of the collision as was proposed by Ramos (1984) and could have probably started in the late Early or Middle Devonian.

The "Autochthonous" Hypothesis

This hypothesis suggests that no important relative displacements occurred between Patagonia and the Brazilian Craton during the Phanerozoic. In this case both SG1 and SG2 must be representative of South America and the whole of Gondwana. The position of SG2 agrees with this hypothesis, showing no significant latitudinal displacement between Patagonia and South America after the Early Devonian, giving additional support to a southern South America position of the South pole during the Late Silurian–Early Devonian. SG1 disagrees with the "loop" path mentioned above, but it agrees surprisingly well with the "reversed polarity" a.p.w.p. proposed many years ago by Schmidt and Morris (1977). As shown in Fig. 14, SG1 is situated half-way between the reversed Ordovician poles and the Late Silurian–Early Devonian poles in the path proposed by Schmidt and Morris (see Fig. 1 of that paper). This would imply a rapid shift of the whole of Gondwana (not only Patagonia) between the Late Ordovician and the end of Silurian. If SG1 actually represents the Wenlockian (425 Ma) Gondwana pole position, there arises an incompatibility with the age of the AIR pole (435 Ma), which is a high-quality paleomagnetic pole. This incompatibility could be solved if we assume a younger

magnetic age for it. A magnetic reset of the rocks from which the AIR pole was obtained in the Late Silurian could have been possible as there are many Late Silurian magmatic rocks in the area (Hargraves *et al.*, 1987). But in this case, one should disregard the K/Ar datings obtained from some paleomagnetic samples indicating an Early Silurian age (435 Ma).

A recent, yet unpublished, paleomagnetic study of the Silurian–Devonian Merenee Sandstone from central Australia (Li *et al.*, 1990, quoted in Schmidt and Embleton, in press), has apparently yielded a paleomagnetic pole close to SG1 (267°E, 55°S), when both are rotated to the Gondwana framework. If this pole is confirmed, it would probably support the reversed polarity path for Gondwana and therefore the autochthonous hypothesis for Patagonia. It could be argued that SG1 (and the unpublished Australian pole, as well) could also support a modified (and very long) "normal" loop path going from northern Africa up to the middle of the Pacific (pole SG1) in the Early Silurian, and coming back to southern South America in the Late Silurian. This would imply a new path for Gondwana on the basis of only one pole, and it also implies an even faster displacement of this supercontinent than that resulting from the "reversed path." This makes this possibility much less probable.

In any case, this interpretation suggests the absence of a stationary Gondwana, with the pole located in southern South America, during the mid–Paleozoic, but a nearly continuous displacement between the Late Ordovician and the Late Devonian.

The paleogeographic implications of the accepted "loop" path and the reversed polarity path are completely different for southern South America in the Silurian. While the first implies very high latitudes during the entire period, the other indicates tropical latitudes during most of it. It is obvious that different stratigraphic features and paleontologic assemblages must be related to each paleogeographic reconstruction; those based only upon the recordings of tillites seem to be no accurate enough (Van Houten and Hargraves, 1987). More investigations in paleoclimatology and biogeography in southern South America, and especially many more reliable paleomagnetic studies in Silurian rocks of Gondwana, are needed to define the APW path of this supercontinent.

CONCLUSIONS

A preliminary paleomagnetic study of the Silurian–Devonian Sierra Grande Formation, exposed in northeastern Patagonia (Argentina), yielded two quite reliable (although preliminary) paleomagnetic poles. These are the first middle Paleozoic poles from Patagonia. One, called SG1, belongs to the Rosales ferriferous horizon, which is intercalated in the lower member of this formation. It is situated at 238.0°E, 3.4°N (D1=14.3°, D2=8.5°) and has been assigned to the Wenlockian. The other (SG2) belongs to the Late Silurian–Early Devonian upper member of the formation and is situated at 283.5°E, 42°S (D1=20.4°, D2=20.2°).

Paleomagnetic data from other rocks, not ferriferous, of the lower member suggest a post–tectonic remagnetization, probably during the Permian. Finally, paleomagnetic data from some Jurassic rhyolitic dikes (Marifil Formation) are consistent with previous data from this formation. They also indicate that the intrusion of these dikes has not affected the magnetizations of the middle Paleozoic rocks.

The positions of SG1 and SG2 indicate a rapid displacement of Patagonia from tropical latitudes in the Wenlockian (425 Ma) to polar latitudes in the Early Devonian (400 Ma). The tectonic implications of the SG1 and SG2 positions are interpreted according to two different hypotheses. One takes into account the previous hypothesis (Ramos, 1984) that proposes that Patagonia collided with the Brazilian Craton during the Paleozoic. According to this, the position of SG1 far from the presently most accepted Silurian–Devonian pole position of Gondwana (Van der Voo, 1988) suggests that in the Wenlockian, Patagonia was situated far from the Gondwanic continents. Nevertheless, the position of SG2 consistent with the Gondwana poles indicates that in the Early Devonian, Patagonia had already collided or, more probably, was next to collide with South America. The other interpretation is based upon the assumption that Patagonia and South America did not undergo important relative movements in the Phanerozoic. According to this tectonic framework, SG1 and SG2 would be representative of the whole of Gondwana. The position of SG2 is consistent with other Late Silurian–Early Devonian poles of Gondwana located in southern South America. On the other hand, the position of SG1 would imply that the AIR pole (Hargraves *et al.*, 1987) is younger, probably Late Silurian, and that the reversed polarity option a.p.w.p. first proposed by Schmidt and Morris (1977) should be taken as the most probable early–middle Paleozoic a.p.w.p. of Gondwana.

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